📘 AI/ML Internship – Day 2 Notes & Tasks

🟢 Module 3: Python for AI (Basics – Part 1)

📌 Part 1: Python Setup for AI

🔹 What is Python for AI?

Python is the most widely used programming language in AI because it allows easy data handling, model building, and experimentation.

🔹 Installing Python

Steps:

1. Download Python from official website
2. Install and check:

python --version

🔹 IDE (Code Editor)

Recommended:

- VS Code
- Jupyter Notebook (for AI later)

📌 Part 2: Python Fundamentals

🔹 Variables

Variables store data values.

```
x = 10
name = "AI Student"
```

🔹 Data Types

- Integer → 10
- Float → 3.14

- String → "Hello"
- Boolean → True / False

🔹 **Type Checking**

```
x = 10
print(type(x))
```

📌 **Part 3: Input & Output**

```
name = input("Enter your name: ")
print("Welcome", name)
```

📌 **Part 4: Operators**

🔹 **Arithmetic Operators**

```
a = 10
b = 5

print(a + b)
print(a * b)
```

🔹 **Comparison Operators**

```
print(a > b)
print(a == b)
```

📌 **Part 5: Conditional Statements**

```
num = 10

if num > 0:
    print("Positive")
else:
    print("Negative")
```

📌 Part 6: Loops

🔹 For Loop

```
for i in range(1, 6):
    print(i)
```

🔹 While Loop

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

📌 Part 7: Functions

```
def add(a, b):
    return a + b

print(add(5, 3))
```

🎯 Tasks for Day 2 (More Practical Focus)

🧠 Task 1: Theory (Short Answers)

1. Why Python is used in AI?
2. What are variables?
3. What is a data type?
4. Difference between if and loop
5. What is a function?

💻 Task 2: Basic Coding (Must Do)

Write programs:

1. Print your details (name, age, course)
2. Add, subtract, multiply two numbers

3. Check even or odd
4. Find largest of two numbers
5. Take input and display it

💻 **Task 3: Logic Building (Important)**

1. Check whether number is positive, negative or zero
2. Find largest of 3 numbers
3. Grade system:
    - Marks > 90 → A
    - 70–90 → B
    - < 70 → C

💻 **Task 4: Loop Practice (Core AI Skill)**

1. Print numbers 1–20
2. Print even numbers
3. Print sum of first N numbers
4. Reverse numbers from 10 to 1

💻 **Task 5: Function Practice**

1. Function to add two numbers
2. Function to check prime number
3. Function to find factorial

⌨️ **Task 6: Git Integration**

1. Create file:

day2.py

2. Add all programs
3. Push to GitHub:

```
git add .
git commit -m "Day 2 Python basics completed"
git push
```

🧩 Task 9: Bonus

1. Print pattern:

```
*
**
***
****
```

2. Reverse a number
3. Count digits in a number